KUNCHALA SHIVA KIRAN

AI/ML Engineer | Full-Stack Developer | M.Tech Computer Science & Engineering

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Warangal, Telangana, India

PROFESSIONAL SUMMARY

Motivated M.Tech CSE student with a proven track record of building production-ready AI/ML systems and full-stack web applications. Designed and deployed two advanced AI research projects — an Explainable AI ransomware detection system achieving **100% accuracy** on the CIC-Ransomware-2019 dataset, and a privacy-preserving federated learning platform simulating real-world GDPR/HIPAA-compliant healthcare AI. Proficient in Python, Java, scikit-learn, Flask, Spring Boot, and cloud deployment. Passionate about bridging the gap between complex AI models and interpretable security solutions.

EDUCATION

Master of Technology (M.Tech) — Computer Science & Engineering — **2025 – 2027 (Ongoing)**
SR University, Warangal, Telangana — Percentage: 78%

Bachelor of Technology (B.Tech) — Computer Science & Engineering — **2020 – 2024**
SR University, Warangal, Telangana — Percentage: 72%

Intermediate (MPC) — Telangana State Board — **2018 – 2020**
SR Junior College, Warangal — Percentage: 66%

SSC — Secondary School Certificate — **2017 – 2018**
Tejaswi High School, Telangana — Percentage: 87%

TECHNICAL SKILLS

Languages: Python, Java (J2EE), JavaScript (ES6+), SQL, HTML5, CSS3

Frameworks: Flask, Spring Boot, Spring Data JPA, React.js, Streamlit, Bootstrap

Tools & Platforms: Git/GitHub, Docker, Render, Hugging Face Spaces, Chart.js, RESTful APIs, Maven

AI/ML: Scikit-learn, Isolation Forest, Random Forest, LIME, Federated Learning, LoRA, Opacus DP-SGD

GenAI & LLMs: Microsoft Phi-3, Hugging Face, LoRA Fine-tuning, Differential Privacy

Core Concepts: Data Structures & Algorithms, OOP, DBMS, SDLC, Agile, XAI, GDPR/HIPAA

KEY PROJECTS

Explainable AI (XAI) for Ransomware Detection — *Python · Scikit-learn · LIME · Flask · Chart.js*

GitHub | Live Demo

- Designed a **Dual-Model XAI Framework** combining **Isolation Forest** and **Random Forest**, achieving **100% accuracy** on the CIC-Ransomware-2019 benchmark dataset.
- Integrated **LIME** to generate transparent justifications for predictions, transforming black-box ML outputs into actionable security audit reports.
- Built a **Flask web application** with a real-time threat visualization dashboard using **Chart.js**.

SecureEdge-GPT | Privacy-Preserving Federated AI System — *Python · Flower · Phi-3 · LoRA · Opacus · Docker*

GitHub | Live Demo

- Built a distributed AI training system for healthcare using **Flower** framework to train **Microsoft Phi-3** without sharing raw patient data, ensuring full **GDPR/HIPAA compliance**.
- Implemented **Adaptive Differential Privacy** (Opacus) and used **LoRA fine-tuning** to reduce model update size from **15 GB to under 100 MB**.
- Deployed as a containerized solution with a real-time **Streamlit dashboard** on Hugging Face Spaces.

JavaCareerHub | Full-Stack Career Advancement Platform — *Java 17 · Spring Boot 3.2 · React 18 · MySQL · Chart.js*

GitHub

- Architected a career platform using **Spring Boot 3.2** and **React 18**, integrating 7+ tools including a resume builder and tech trend dashboard.
- Designed **RESTful APIs** and implemented **JWT authentication** for secure user access.

Profile Analyzer Through Smart Glasses — *Python · Computer Vision · Face Recognition*

- Developed a computer vision system enabling real-time **facial recognition** for instant student data retrieval.

CERTIFICATIONS & ACHIEVEMENTS

- **AWS Cloud Foundations** — Amazon Web Services (AWS) | March 2025
- **Cybersecurity Essentials** — CISCO Networking Academy | February 2023
- **Full-Stack Developer** — Deployed 3+ professional-grade web applications with live users.
- **Open Source Contributor** — Active presence on GitHub with 10+ public research repositories.